Air Transat Flight 236

Flight model: Airbus A-330
Date: 24 August, 2001

From: Tonado ,Canada
To: Lisbon, Portuga

The Airbus A330 is a twin engine plane. The scheduled path for the plane is over Atlantic ocean.

Problem

At the Dark. The flight took off normal. To avoid weather related problems the plane was diverted off the pre-planed path. Which is a little longer than the initial path. In the airbus A330, the pilots have to keep an eye on the fuel consumed by the plane by performing manual calculations. Then, there is a warning which raised to indicate the oil temperature is low in the second engine but the oil pressure is high in the same. After a couple of minutes a fuel imbalance warning is raised. The pilots plans to moves the fuel through a **cross feed valve**. But in reality, there is a fuel leak in the second engine. Despite of multiple warning, the pilots believed that it many be mainly dues to computer errors. After while, the co-pilot finds that the is a possibility of a fuel leak. The pilots ask the air-hostess to perform a visial check to find for any traces for fuel leak. In case of a fuel leak, there must be a trail behind the engine made up of fuel droplets. The pilots declare fuel emergency. After a hour from the first warning, the right engine runs out of fuel. The pilots increase the engine power on the left engine to manage the imbalance. The one engine is not enough to fly at high altitude and must descend immediately. The pilots still believe a problem with the flight computer and would be rectified soon. The pilots plans to land the plane in a alternative airport. Later both engine losses fuel and fails. The plane then starts to glide over the Atlantic. The plane losses power to operate the electrical systems. The cabin soon starts depressurizing without power. The plane does not have enough fuel to reach the nearest airport. At the same time, the modern jets are not designed to float upon clash landing into the water dues to a now tail shape. The pilots were able to bring the plane near glide towards the airport. Near the runway the plane speed is higher than usual. The pilots perform maneuvers to lower the speed. Despite the best efforts, the speed havent decrease to the require limit. The crew only has speed breaks, all other breaking mechanism failed to operate due to lack of power. At the end after applying the breaks the plane was able to stop the plane on the runway.

Investigation

Right before the flight, the engines had a full maintenance and were fitted with a new engine. But the engine manufacture had not supplied with one system required for its functioning and the ground crew had tried to fit an old system with the new engine which made the fuel pipeline to rupture. The pilots had failed to shut of the cross feed value despite identifying a potential fuel leak. According to the pilots, they did what the flight manual had instructed them to do. But, they failed to analyses if they had achieved the result that they needed was achieved by performing the action.

Changes that were brought after the incident

Airbus now have the compute system to calculate the fuel needed with respect to the flight plan and alert the pilots if more fuel is being consumed or lost than what is actually needed.

Reference

1. Airbus A330 Runs Out of Fuel Over Atlantic Ocean — Mayday: Air Disaster
   https://www.youtube.com/watch?v=-Os3DBCd-Ws&list=WL&index=2